

Effect of photon quantum statistics on electrons in above-threshold ionization

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Strong field physics and quantum optics, two prosperous fields in modern science, have recently been connected with each other. Here, we study the quantum effect of bright squeezed vacuum light and thermal light on above-threshold ionization. With quantum strong-field approximation theory, we reveal a decoherence phenomenon caused by the statistics of photons, in which above-threshold ionization could be degenerated into a classical-like picture under the effect of quantum light. This represents a direct transition from the wavelike nature to the particlelike behavior of photoelectrons. We also reveal the ionization enhancement effect of the bunched light in photoionization from the perspective of autocorrelation function $g^{(n)}$. The number distributions of electrons ionized by different quantum light fields are studied, and it is shown that the photon number distribution can be mapped onto the photoelectron number distribution. This finding enables the production of bunched electrons with strong-field ionization with quantum light.

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Introduction. Attosecond science and strong field laser physics open the door for studying nonequilibrium dynamics of atomic system in electron intrinsic timescale. There, above-threshold ionization (ATI) [1,2], and high-harmonic generation (HHG) [3,4] stand out as two pivotal processes. The discovery of these phenomena has ushered people into the realm of attosecond measurements [5] assisted by semiclassical theories such as strong field approximation (SFA) which treats atoms quantum-mechanically but treats electromagnetic fields classically [6]. In recent years, the quantum light sources with high intensity are becoming experimentally obtainable, making it possible to measure quantum optical effects in strong field physics [7–9]. Recent studies with fully quantized approach have revealed the existence of numerous novel phenomena in intense light-matter interaction, such as the modification of ionization rates [10,11] and harmonic yields [12,13], the generation of photon entangled coherent states [14,15], and squeezed optical states [16]. Quantum effects of the light field influence the motion of electrons in attosecond time scale [13,17]. These effects are all overlooked in semiclassical theories. The quantized study of intense-light-matter interaction serves as a bridge connecting strong field physics and quantum optics, making strong field physics as testing ground for foundational quantum physics and a platform for quantum information [18,19].

Squeezed states of light represent a typical form of quantum optical fields. They are states satisfying minimal uncertainty, characterized by decreased fluctuations in one quadrature and increased fluctuations in the other compared to coherent or vacuum states. In quantum optics, the generalized Glauber P representation forms a correspondence between the quantum and the classical coherence theory [20]. Based on this premise, quantum strong-field approximation (qSFA) theory has been developed to describe HHG [12,13] and ATI [10] driven by quantum light. Recently, a one-dimensional solution of time-dependent Schrödinger equation was used to investigate strong-field ionization by quantum light [11]. However, to distinguish the quantum light from classical optical fields, the role of autocorrelation in strong-field ionization has yet to be explored. Depending on whether the second-order autocorrelation function $g^{(2)} > 1$ or $g^{(2)} < 1$, a light field is referred to as “bunched light” or “antibunched light.” In the perturbative regime, such as low-order harmonics [21] and multiphoton ionization [7], the property of bunching can enhance the n th order multiphoton process and the enhancement factor is equal to $g^{(n)}$. For the bright squeezed vacuum (BSV) and thermal states, the enhancement factors are $(2n - 1)!!$ and $n!$, respectively. As the intensity of squeezed vacuum light increases [22], it would be interesting to explore the impact of bunched light on the ionization rates and photoelectron momentum distribution of ATI, which is in a nonperturbative regime. On the other hand, the autocorrelation function is directly related with the photon number distribution of the driven light. Hence, it would also be interesting to study the number distribution of generated photoelectrons in the strong-field regime.

In this paper, we study the quantum optical nature of ATI of hydrogen atoms with two kinds of bunched lights, i.e., BSV and thermal light, which respectively represent typical

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examples of quantum light, and compare the results with those obtained using coherent light. In the framework of the qSFA theory, we show that the bunched lights not only enhance the ionization rates of photoelectrons, but also alters its statistics of momentum. Besides, the broad photon number distribution in bunched lights leads to the decoherence of photoelectrons and smears out its interference patterns in momentum space. This directly manifests the particle nature of photoelectrons in strong-field ionization, which shows a sharp contrast to the wavelike properties exhibited by electrons in momentum space in nearly all previous studies of ATI. Moreover, we explore the number distribution of generated electrons and investigate its relationship with the photon number distribution of intense driving light. In this way, we show that the quantum statistics of photons are projected on electrons through ATI.

Theoretical methods. We calculate the photoelectron momentum distributions with quantum light using qSFA theory. Keldysh–Faisal–Reiss theory or SFA theory is one of the corner stones for studying strong-field physics [23–25], which has been extended for the study with quantum light [10,12,13]. Within the framework of qSFA theory, the probability amplitude of photoelectron at final momentum $\mathbf{p}$ is given by (see the Appendix)

$$W(\mathbf{p}) = \int dE_\alpha P(E_\alpha) |M_\alpha(\mathbf{p})|^2, \quad (1)$$

where $M_\alpha(\mathbf{p})$ represent the probability amplitude corresponding to the driving coherent light field with amplitude E_α . $P(E_\alpha)$ is the quasiprobability distribution characterizing the statistics properties of the quantum optical fields. In order to explore the influence of the statistics of light on ATI, we begin by simulating the photoelectron momentum distributions driven by BSV, thermal, and coherent light, which respectively represent typical examples of light in quantum optics. To be specific [Fig. 1(c)],

$$P(E_\alpha) = \begin{cases} \sqrt{\frac{1}{2\pi E_0^2}} \exp\left(-\frac{E_\alpha^2}{2E_0^2}\right) & \text{(BSV state)} \\ \frac{1}{\sqrt{\pi E_0^2}} \exp\left(-\frac{E_\alpha^2}{E_0^2}\right) & \text{(thermal state)} \\ \delta(E - E_\alpha) & \text{(coherent).} \end{cases} \quad (2)$$

As is well known, the coherent light is a classical light field with a Poissonian photon number distribution, while the thermal light exhibits super-Poissonian distribution and BSV reveals an even more broad super-Poissonian distribution. Besides, BSV light represents a typical example of nonclassical macroscopic state of light. In our simulation, we choose hydrogen atoms as the target benchmark. We maintain the electric field strength at $E_0 = 0.025$.

Results and discussion.

Photoelectron momentum distribution. We have calculated photoelectron momentum distributions for photoionization of Hydrogen atom with BSV, thermal, and coherent light at the wavelength of 500 nm and 1000 nm, as shown in Fig. 2. In the calculation, we use a linear polarized pulse with the form $E(t) = E_\alpha f(t) \sin(\omega t)$, with $f(t)$ the cosine-squared envelope with a full duration of six optical cycles. It turns out that their interference patterns of ionized electrons are very different (explained later), indicating that the electron momentum spectra in ATI is influenced by the quantum statistics of the driving field.

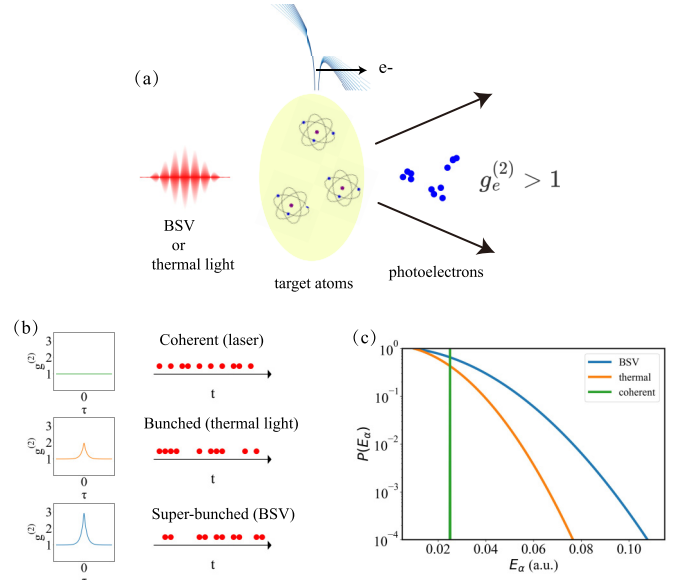


FIG. 1. (a) The bunching characteristics of quantum light fields will be transferred to photoelectrons through the ionization process. For a light field with $g^{(2)} > 1$, the electrons ionized by this light field will also have $g_e^{(2)} > 1$. (b) The distribution of $g^{(2)}(\tau)$ for three types of light fields. Coherent light exhibits Poissonian statistics, where photons are emitted independently. The second-order correlation function $g^{(2)}(0) = 1$. Thermal light shows significant photon bunching, leading to $g^{(2)} = 2$. BSV exhibits even stronger photon bunching, where photons tend to appear in pairs. This lead to $g^{(2)} = 3$. (c) The corresponding quasiprobability distribution of electric field amplitude.

Besides, one can see that, for the long wavelength of driving laser, the intracycle and intercycle interference [26,27] in final electron spectra becomes blurry. In the case of 1000 nm driving fields, these interference patterns are totally smeared out [Figs. 2(d) and 2(e)]. Such a decoherence phenomenon of electron wavepackets is unique for ATI in quantum light field. Interference can be considered as the fundamental property of matter waves in quantum mechanics. It implies that as the degree of decoherence increases, the appearance of photoelectrons in the final state exhibits a transition from a quantum picture (wavelike) to a classical picture (particlelike).

The decoherence is associated with the broad photon number distribution. This phenomenon also exists in other quantum systems, such as free-electron-photon interactions [28], Rabi oscillation [29], and Compton scattering [30]. For the BSV and thermal light, the photon number distribution per pulse can be expressed as [Fig. 3(a)]

$$P(N_\alpha) = \begin{cases} \frac{1}{\sqrt{2\pi N_0 N_\alpha}} e^{-\frac{N_\alpha}{2N_0}} & \text{(BSV)} \\ \frac{1}{N_0} e^{-\frac{N_\alpha}{N_0}} & \text{(thermal state),} \end{cases} \quad (3)$$

where N_α is the photon number and N_0 is the average photon number per pulse. According to $N_0 \propto E_0^2/\hbar\omega$, keeping E_0 constant while decreasing the frequency is equivalent to increasing the statistical broadness of the photon number distribution. This explains why the 1000 nm case exhibits the maximum degree of decoherence in Fig. 2.

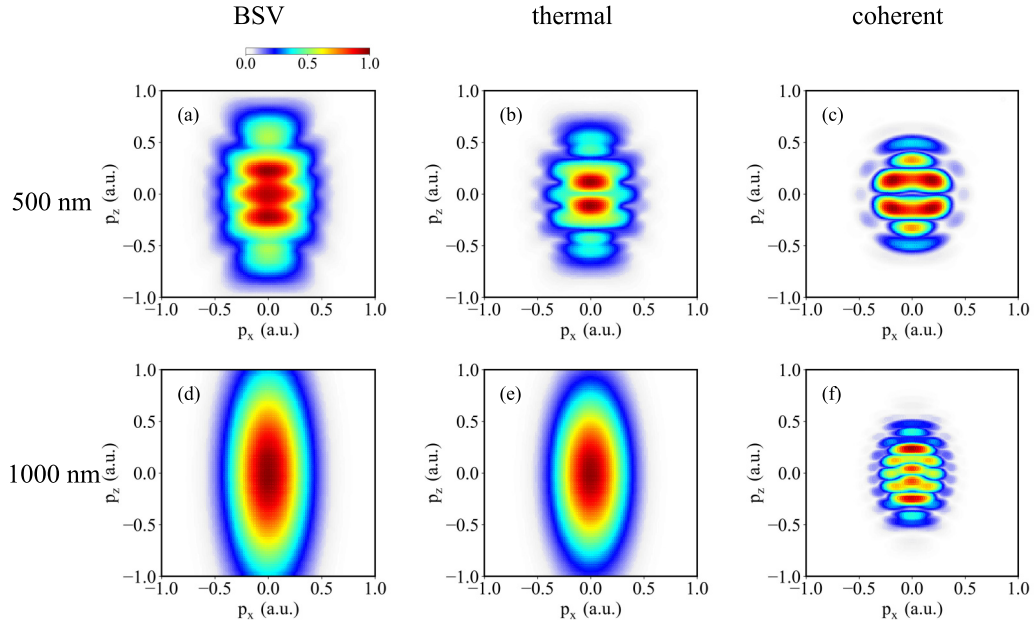


FIG. 2. Photoelectron momentum distribution ionized by (a), (d) BSV; (b), (e) thermal; and (c), (f) coherent state. (a)–(c) correspond to a wavelength of 500 nm, while (d)–(f) correspond to a wavelength of 1000 nm. The electric field strength is fixed at $E_0 = 0.025$ a.u.

The broad photon number distribution also influences the motion of electrons in the intense driving field, reflected in the ponderomotive energy [17]. In a classical electric field, U_p can be expressed as $U_p = E_0^2/4\omega^2$, where E_0 is the peak amplitude and ω is the frequency of electric field. In Fig. 3(a), we show the statistical properties of the quantum light. We can rewrite ponderomotive energy as $U_p = E_\alpha^2/4\omega^2 \propto I_\alpha \lambda^2$,

where λ is the wavelength of driving field. Here we have used the relationship $I_\alpha \propto E_\alpha^2$. In the tunneling ionization regime, due to the AC stark effect, the final energy of ionized electron is $E_k = n\hbar\omega - I_p - U_p$ [31]. And thus, the position of the ATI peak will change accordingly. According to Eq. (1), the final momentum distribution is the statistical result of the driving light; it therefore leads to the blurring of the

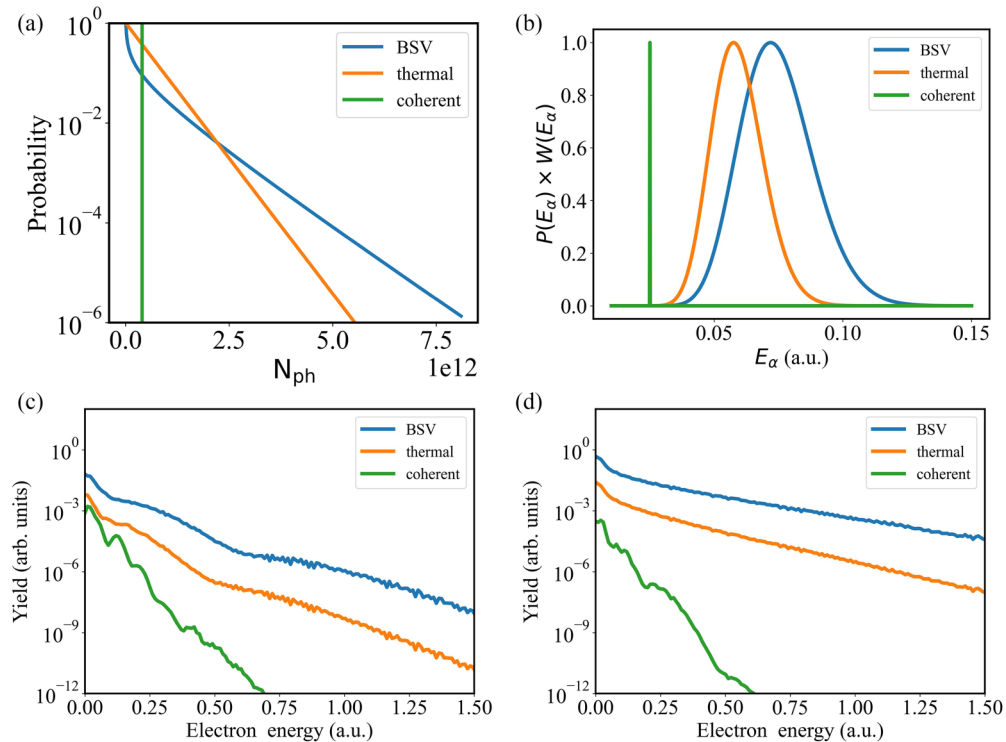


FIG. 3. (a) Photon number statistics of BSV, thermal, and coherent light. (b) Ionization probability of BSV, thermal, and coherent light. Photoelectron energy spectra under driving light wavelengths: (c) 500 nm (d) 1000 nm.

momentum interference patterns. Moreover, since $U_p \propto I_a \lambda^2$, for the longer wavelength, it has greater variation in the ATI peak. Consequently, it has larger effect of decoherence.

Photoelectron energy spectra. Besides, as seen in Figs. 3(c) and 3(d), one can observe that the ionization probability for the BSV and thermal state are significantly higher than that for coherent state. We can interpret this phenomenon from two perspectives. On one hand, this can be understood from the bunching nature of the light field. The degree of bunching of light is characterized by the n th order autocorrelation function, which is defined as $g^{(n)} = \langle :N^n: \rangle / \langle N \rangle^n$ [32,33]. Here, $\langle :N^n: \rangle = \langle (a^\dagger)^n a^n \rangle$ is the normally ordered n th order moment of the photon number. The n -order autocorrelation function of BSV and thermal light is $g^{(n)} = (2n-1)!!$ and $g^{(n)} = n!$, respectively. Although in the nonperturbative regime, ATI remains a multiphoton process, which is still enhanced by the bunching property. Since the autocorrelation function of BSV state is larger than that of the thermal state, which we call “superbunching” [34], the ionization yield is correspondingly larger. On the other hand, we can comprehend this phenomenon from the perspective of electric field fluctuation. As shown in Eq. (2), unlike coherent state, which has a fixed electric field strength, the electric field distribution of thermal states follows the Gaussian distributions. Furthermore, that of BSV exhibits a broader distribution compared to the thermal state. Due to the fact that the ionization contributions from BSV and thermal light predominantly arise from high electric field strength [Fig. 3(b)], tunneling ionization becomes the dominant ionization mechanism under such conditions. To analyze the statistical properties of electrons, we utilize the ionization rate given by the Ammosov-Delone-Krainov (ADK) theory [23,35–37] $W(E_\alpha) = \frac{2e^2}{\pi} \left(\frac{3}{\pi E_\alpha} \right)^{1/2} \exp\left(-\frac{2}{3E_\alpha}\right)$. Here atomic units are used. By multiplying the probability distribution of the electric field with the ionization rates, we obtain the probability distribution for ionization rates. One can clearly observe that the peak positions for the three types of light are in descending order for BSV, thermal, and coherent light. The peak position of the probability density can be regarded as the equivalent electric field strength for each type of light field. With this, one can understand the phenomenon of ionization enhancement by the BSV light.

Photoelectron number distribution. Next, we explore the statistics of the electrons generated from ATI by different forms of light, which is overlooked in semiclassical theories [38]. The probability distribution of photoelectron $P_e(N_e)$ can be obtained from the driven field as $P_e(N_e)dN_e = P(N_\alpha)dN_\alpha$, where N_e represents the photoelectron number. By using $N_\alpha = K_1 E_\alpha^2$, we rewrite ionization rate Eq. (4) as a function of the driving photon number,

$$W(N_\alpha) = \frac{2\sqrt{3}}{\pi^{3/2}} \left(\frac{K_1}{N_\alpha} \right)^{1/4} \exp\left(-\frac{2K_1^{1/2}}{3N_\alpha^{1/2}}\right), \quad (4)$$

where K_1 depends on the laser parameters. To be specific, N_α can be expressed approximately by $N_\alpha \approx I_a \tau S / \hbar \omega = K_1 E_\alpha^2$, where τ is the duration of the laser pulse and S is the laser focus size. Here we take $\tau = 100$ fs and $S = 10^{-8}$ m², leading to the value of $K_1 \approx 10^{15}$. Next we can express the electron number as $N_e = K_2 W(N_\alpha)$, where K_2 represents the number of atoms interacting with the laser field and mainly

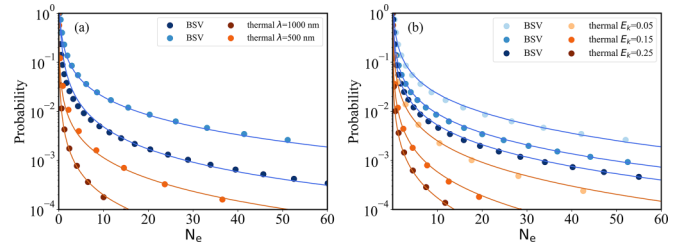


FIG. 4. (a) The probability distribution of electron number for driving light wavelength of 500 nm and 1000 nm. (b) The electron number distribution for final energy of 0.05 a.u., 0.15 a.u., and 0.25 a.u. with driving wavelength of 500 nm. The dots are obtained from the ADK theory and the lines are fitted using the generalized Gamma distribution function. The probability has been normalized.

depends on the experimental conditions such as the focus size of laser beam and gas density. Here we take $K_2 = 1 \times 10^7$. It is worth noting that the specific values of K_1 and K_2 do not affect the main conclusions drawn in our study. Solving the inverse function of this expression, we obtain

$$N_\alpha = 16K_1/9 * \text{lambertw}\left(-\pi^3 N_e^2 / 9K_2^2\right). \quad (5)$$

Here $\text{lambertw}(\cdot)$ represents the Lambert W function [39], which is the inverse function of $y = xe^x$. It is worth noting that we are considering the tunneling ionization process here, which is a nonperturbative process and distinct from perturbative ionization which obeys $N_e = KN_\alpha^n$. Up to this point, we can obtain the number distribution of photoelectrons as

$$P_e(N_e) = P_\alpha(N_\alpha) \frac{dN_\alpha}{dN_e}. \quad (6)$$

By applying Eqs. (5) and (6), we extract the electron number distribution as depicted in Fig. 4(a). We can see that, unlike the deterministic electron emission associated with coherent light, the probability of electrons ionized by the BSV and thermal light exhibit a heavy-tail distribution. This actually reflects that the bunching characteristics of photons are transferred to electrons through ionization. In other words, the bunched and super-bunched electrons are generated with the thermal light and BSV light, respectively.

Above, we have discussed the distribution of total number of photoelectrons. During the ATI process, we know that electrons can absorb different numbers of photons, resulting in series energy peak (ATI peak). Therefore, one should explore the number distribution of electrons with different final kinetic energy. Within the tunneling ionization picture, the energy spectrum of photoelectrons can be analytically obtained as [37]

$$W(E_\alpha) = \frac{4\omega^2}{\pi^2 p E_\alpha^2} \exp\left(-\frac{2}{3E_\alpha} - \frac{p^2 \gamma^3}{3\omega}\right), \quad (7)$$

where ω is the frequency of driving field, $\gamma = \omega/E_\alpha$ is the Keldysh parameter, $p = \sqrt{2E_k}$ is the final momentum of electron, and E_k is the kinetic energy. In Fig. 4(b), we show the analyzed number distribution of electrons with different energy at 1000 nm for the BSV and thermal light. The dots in the figure are obtained using Eqs. (7) and (8), following the same procedure as obtaining Fig. 4(a). To delve deeper into the distribution of electrons at different energies,

we employ the generalized gamma function to fit the data obtained through the ADK tunneling formula [Eq. (8)]. The generalized n th gamma function is given by

$$P(N_e) = \begin{cases} \frac{2^{n/2} \sqrt{(2n-1)!!}}{n \sqrt{2\pi}} \frac{2^{n/2} \sqrt{N_0} N_e^{1-1/2n}}{e^{-\frac{1}{2} \sqrt{(2n-1)!!} \frac{N_e}{N_0}}} & \text{(BSV)} \\ \frac{n!}{n N_0 N_e^{1-1/n}} e^{-\sqrt{n!} \frac{N_e}{N_0}} & \text{(thermal state),} \end{cases} \quad (8)$$

where N_0 represents the average electron number. Here N_0 has considered the ionization enhancement effect characterizing by $g^{(n)}$ [22]. As shown in Fig. 4(b), for the BSV light at 1000 nm, the electrons with final energies of $E_k = 0.05, 0.15, 0.25$ a.u. correspond to average electron numbers of $N_0 = 11.0, 10.0, 8.0$. The effective orders are $n = 2.4, 2.9, 3$, respectively. For thermal light, these values are $N_0 = 4.0, 1.8, 1$ and $n = 3.1, 3.2, 3.4$, respectively. This conforms to the trend that higher energies correspond to higher orders of the ionization process, resulting in the decreasing of the average number of photoelectrons. By analogy, with the definition of second-order coherence $g^{(2)}$ for quantum light, we can define the second-order coherence $g_e^{(2)}$ for electrons. It can be extracted through the electron number distribution: $g_e^{(2)} = 1 + \frac{(\Delta n)^2 - \langle n \rangle}{\langle n \rangle^2}$. Here Δn is the standard variance σ and $\langle n \rangle$ is the average electron number. For the n th-order multiphoton ionization process driven by BSV and thermal light, $g_e^{(2)}$ can be obtained from Eq. (8). The results are $(2n-1)!!$ and $n!$, respectively. For tunneling ionization, an effective n can be used as discussed earlier. Bunched electrons are typically used in scenarios such as free-electron lasers, where they are modulated by electromagnetic fields [40]. Here, we show that using the quantum nature of light to directly generate bunched electrons, offering an approach for modulating the quantum optical nature of electrons through photoionization.

Conclusion. In conclusion, we have studied the effect of quantum statistics of light on above-threshold ionization. We show that the broad photon number distribution of quantum light leads to the decoherence of photoelectrons in momentum space. This enables us to modulate the transition of electrons from wavelike to particlelike behavior by altering the quantum statistics of light. We show that the bunching nature of BSV and thermal light enhances the photoionization with the same average power as coherent light, which implies that it might also generally enhance the interaction between light and matters. Besides, we find that unlike the classical light field with a certain ionization probability, the photoelectron ionization probability in quantum light exhibits a broad number distribution. This provides a potential opportunity to generate free electron source with a nontrivial electron number distribution by controlling the photon number distribution of driving light [41]. The photoelectrons can serve as nondestructive detection of the state of the light [42]. The generation of bunched and superbunched photoelectrons with quantum light are also potentially the electron sources of accelerators and free-electron lasers.

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Appendix: Framework of the quantum strong-field approximation. We start from the Schrödinger equation for the evolution of the density matrix:

$$i\hbar \frac{\partial \rho_0(t)}{\partial t} = [\hat{H}, \rho_0(t)], \quad (A1)$$

where the Hamiltonian is given by $\hat{H} = \hat{H}_F + \hat{H}_A + \hat{\mathbf{d}} \cdot \hat{\mathbf{E}}$, with $\hat{H}_A = \frac{\hat{p}^2}{2m} + V(\hat{r})$, representing the Hamiltonian of a bound electron, and $\hat{\mathbf{d}}$ is the dipole moment operator, typically defined as $\hat{\mathbf{d}} = -e\hat{\mathbf{r}}$. The Hamiltonian of the free electromagnetic field $\hat{H}_F$ is given by $\hat{H}_F = \sum_{\mathbf{k}, \sigma} \hbar \omega_{\mathbf{k}} (\hat{a}_{\mathbf{k}, \sigma}^\dagger \hat{a}_{\mathbf{k}, \sigma} + \frac{1}{2})$. For simplicity, we consider here the driving field in a single mode: $\hat{H}_F = \hbar \omega (\hat{a}^\dagger \hat{a} + \frac{1}{2})$. The electric field operator is given by $\hat{\mathbf{E}} = i\epsilon^{(1)} \hat{\mathbf{e}} (\hat{a} - \hat{a}^\dagger)$, where $\epsilon^{(1)} = \sqrt{\frac{\hbar \omega}{2V\epsilon_0}}$ represents the amplitude of the electric field and $\hat{\mathbf{e}}$ is the unit polarization vector associated with the mode.

Using the interaction picture transformation $\rho_0(t) \rightarrow e^{-\frac{i\hat{H}_F t}{\hbar}} \rho_0(t) e^{\frac{i\hat{H}_F t}{\hbar}}$, we obtain the following equation:

$$i\hbar \frac{\partial \rho_0(t)}{\partial t} = [\hat{H}_0(t), \rho_0(t)], \quad (A2)$$

where $\hat{H}_0(t) = \hat{H}_A + \hat{\mathbf{d}} \cdot \hat{\mathbf{E}}(t)$ and $\hat{\mathbf{E}}(t) = i\epsilon^{(1)} \hat{\mathbf{e}} (\hat{a} e^{-i\omega t} - \hat{a}^\dagger e^{i\omega t})$. The initial value of the density matrix can be considered as the tensor product of the initial density matrix of the electromagnetic field and the initial density matrix of the atom: $\rho_0(0) = \rho_F(0) \otimes \rho_A(0)$, assuming that the initial states of the electromagnetic field and the atom are independent. According to the Glauber-P representation in quantum optics, the initial density matrix of the electromagnetic field $\rho_F(0)$ can be expressed as a superposition of coherent states:

$$\rho_F(0) = \int d^2\alpha d^2\beta P(\alpha, \beta^*) |\alpha\rangle \langle \beta|, \quad (A3)$$

where $P(\alpha, \beta^*)$ is the Glauber-P function, which serves as a quasiprobability distribution that describes the quantum state of the field in the coherent state basis. It can be used to describe nonclassical light (e.g., squeezed states, photon number states). On the other hand, the atomic system is initially in the ground state $\rho_A(0) = |g\rangle \langle g|$. Thus, the initial density matrix of the entire system can be written as

$$\rho_0(0) = \int d^2\alpha d^2\beta P(\alpha, \beta^*) |\alpha\rangle \langle \beta| \otimes |g\rangle \langle g|. \quad (A4)$$

Here, $P(\alpha, \beta^*)$ is time independent since it does not involve the evolution of the system. Considering that the evolution equation (2) is linear, we can express $\rho_0(t)$ as

$$\rho_0(t) = \int d^2\alpha d^2\beta P(\alpha, \beta^*) \rho_{\alpha\beta}(t), \quad (A5)$$

where $\rho_{\alpha\beta}(t)$ satisfies

$$i\hbar \frac{\partial \rho_{\alpha\beta}(t)}{\partial t} = [\hat{H}_0(t), \rho_{\alpha\beta}(t)] \quad (A6)$$

and $\rho_{\alpha\beta}(0) = |\alpha\rangle \langle \beta| \otimes |g\rangle \langle g|$. The solution can be written as $\rho_{\alpha\beta}(t) = U(t) \rho_{\alpha\beta}(0) U^\dagger(t)$, where $U(t)$ is the time evolution operator, defined as $U(t) = \mathcal{T} \exp(-\frac{i}{\hbar} \int_0^t \hat{H}_0(t') dt')$. Here $\mathcal{T}$ represents the time-ordering operator.

We proceed by using the coherent shift operators $D(\alpha)$ and $D(\beta)$, with coherent parameters α and β , to redefine a matrix

element: $\tilde{\rho}_{\alpha\beta}(t) = D^\dagger(\alpha)\rho_{\alpha\beta}(t)D(\beta)$. Thus yielding $\hat{\rho}_{\alpha\beta}(t) = (D^\dagger(\alpha)U(t)D(\alpha))|g\rangle\langle g| \otimes |0\rangle\langle 0|(D^\dagger(\beta)U(t)D(\beta))^\dagger$. We then define $|\Psi_\alpha(t)\rangle = D^\dagger(\alpha)U(t)D(\alpha)|g\rangle \otimes |0\rangle$, which allows us to express

$$\tilde{\rho}_{\alpha\beta}(t) = |\Psi_\alpha(t)\rangle\langle\Psi_\beta(t)|. \quad (\text{A7})$$

By definition, $|\Psi_{\alpha/\beta}(t)\rangle$ satisfies the Schrödinger equation

$$i\hbar \frac{\partial |\Psi_{\alpha/\beta}(t)\rangle}{\partial t} = H_{\alpha/\beta}(t)|\Psi_{\alpha/\beta}(t)\rangle, \quad (\text{A8})$$

where $|\Psi_{\alpha/\beta}(0)\rangle = |g\rangle \otimes |0\rangle$ and $H_{\alpha/\beta}(t) = \hat{H}_A + \hat{\mathbf{d}} \cdot \mathbf{E}_{\alpha/\beta}(t) + \hat{\mathbf{d}} \cdot \hat{\mathbf{E}}(t)$. The components $E_\alpha(t) = \langle\alpha|\hat{E}|\alpha\rangle$ and $E_\beta(t) = \langle\beta|\hat{E}|\beta\rangle$ correspond to the classical parts of the electric field. Assuming that there is nonentanglement between the atomic system and the electromagnetic field during the interaction process. $\tilde{\rho}_{\alpha\beta}$ can be decomposed as

$$\tilde{\rho}_{\alpha\beta}(t) = |\phi_\alpha(t)\rangle\langle\phi_\beta(t)| \otimes \tilde{\rho}_{\text{light}}. \quad (\text{A9})$$

As we know, $|\phi_{\alpha/\beta}\rangle$ satisfy the semiclassical Schrödinger equation:

$$i\hbar \frac{\partial |\phi_{\alpha/\beta}(t)\rangle}{\partial t} = H_{\alpha/\beta}|\phi_{\alpha/\beta}(t)\rangle, \quad (\text{A10})$$

where $\hat{H}_{\alpha/\beta} = \hat{H}_A + \hat{\mathbf{d}} \cdot \mathbf{E}_{\alpha/\beta}$ is the semiclassical Hamiltonian. In Refs. [12–14], the focus is on the high harmonic generation (HHG) of the light field, and thus the electronic degrees of freedom are traced out. However, since we are considering the ATI process here, we retain the electronic part and trace out the light field, obtaining the electron density matrix as

$$\rho_F(t) = \int d^2\alpha d^2\beta P(\alpha, \beta^*) |\phi_\alpha(t)\rangle\langle\phi_\beta(t)|. \quad (\text{A11})$$

In the ATI process, we aim to obtain the probability for a free electron with momentum $\mathbf{p}$. This probability is given by

$$\begin{aligned} W(\mathbf{p}) &= \text{tr}(|\mathbf{p}\rangle\langle\mathbf{p}| \rho_F(t = \infty)) \\ &= \int d^2\alpha d^2\beta P(\alpha, \beta^*) M_\alpha(\mathbf{p}) M_\beta^*(\mathbf{p}). \end{aligned} \quad (\text{A12})$$

Here $M_\alpha(\mathbf{p}) = \langle\mathbf{p}|\phi_\alpha(t = \infty)\rangle$ represents the probability amplitude for the electron to have momentum $\mathbf{p}$.

Based on whether α is equal to β^* , we can decompose the expression for $W(\mathbf{p})$ into two parts, representing the diagonal

and off-diagonal contributions, respectively,

$$W(\mathbf{p}) = W_{\text{diag}}(\mathbf{p}) + W_{\text{off-diag}}(\mathbf{p}). \quad (\text{A13})$$

Here $W_{\text{diag}}(\mathbf{p}) = \int d^2\alpha P(\alpha, \alpha^*) |M_\alpha(\mathbf{p})|^2$ for $\alpha = \beta^*$ and $W_{\text{off-diag}}(\mathbf{p}) = \int d^2\alpha d^2\beta P(\alpha, \beta^*) M_\alpha(\mathbf{p}) M_\beta^*(\mathbf{p})$ for $\alpha \neq \beta^*$. The diagonal terms represent the incoherent addition of the probability amplitudes and off-diagonal terms represent the interference between different coherent components.

In the main text, we considered only the diagonal terms while ignoring the contribution from the off-diagonal terms. Next, we will justify the validity of this assumption. The relationship between the Glauber-P function $P(\alpha, \beta^*)$ and the Huisi- Q function Q can be expressed as follows:

$$P(\alpha, \beta^*) = \frac{1}{4\pi} \exp\left(-\frac{|\alpha - \beta^*|^2}{4}\right) Q\left(\frac{\alpha + \beta^*}{2}\right). \quad (\text{A14})$$

The coherent state amplitude is related to the electric field as follows:

$$E_\alpha(t) = i\epsilon^{(1)} \hat{\mathbf{e}}(\alpha e^{-i\omega t} - \alpha^* e^{i\omega t}), \quad (\text{A15})$$

where $\epsilon^{(1)} = \sqrt{\frac{\hbar\omega}{2V\epsilon_0}}$ represents the elementary excitation amplitude. We then introduce the complex amplitude of the field as $\mathcal{E}_\alpha = 2\epsilon^{(1)}\alpha$. To proceed further, we consider the limit where $\epsilon^{(1)} \rightarrow 0$ while $\alpha \rightarrow \infty$, such that the complex field amplitude $\mathcal{E}_\alpha$ remains constant throughout this limiting process. Under these circumstances, we obtain

$$\begin{aligned} P(\alpha, \beta^*) d^2\alpha d^2\beta &= \lim_{\epsilon^{(1)} \rightarrow 0} \left(\frac{1}{4\pi} \frac{1}{4|\epsilon^{(1)}|^2} \exp\left(-\frac{|\mathcal{E}_\alpha - \mathcal{E}_\beta|^2}{16|\epsilon^{(1)}|^2}\right) \right) \\ &\times Q\left(\frac{\alpha + \beta}{2}\right) d^2\alpha d^2\beta. \end{aligned} \quad (\text{A16})$$

Given that $Q(\frac{\alpha + \beta}{2})$ is positive and has an integral equal to unity, we can take the limit $\epsilon^{(1)} \rightarrow 0$ separately for the exponential term:

$$\lim_{\epsilon^{(1)} \rightarrow 0} \frac{1}{4\pi} \frac{1}{4|\epsilon^{(1)}|^2} \exp\left(-\frac{|\mathcal{E}_\alpha - \mathcal{E}_\beta|^2}{16|\epsilon^{(1)}|^2}\right) = \delta^{(2)}(\mathcal{E}_\alpha - \mathcal{E}_\beta). \quad (\text{A17})$$

This indicates that the Glauber-P function becomes highly localized, and all its contributions converge to the diagonal terms in the phase space defined by $\mathcal{E}_\alpha = \mathcal{E}_\beta$. For simplicity, and without loss of generality, we consider a real electric field E_α . Thus, we arrive at the key formula used in the main text:

$$W(\mathbf{p}) = \int dE_\alpha P(E_\alpha) |M_\alpha(\mathbf{p})|^2. \quad (\text{A18})$$

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